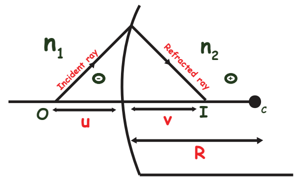
RAY OPTICS -2

REFRACTION AT CURVED SURFACES



u = -ve
V = +ve
R = +ve

$$\frac{n_2}{v} - \frac{n_1}{-u} = \frac{n_2-n_1}{R}$$

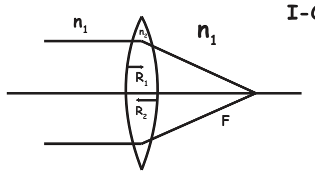
all lengths on the side of incident ray are taken as -ve
all lengths on the side of reflected ray are taken as +ve

I-O=S
$$\frac{\text{Medium}}{\text{I.Distance}} - \frac{\text{Medium}}{\text{O.Distance}} = \frac{\text{Change in medium}}{\text{Radius of curvature}}$$

Transverse Magnification $T.M = m = \dfrac{\frac{v}{n_2}}{\frac{u}{n_1}} = \dfrac{v \times n_1}{u \times n_2}$

LENS MAKERS FORMULA

To find focal length if refractive index is same on both sides of lens.



I-O=S $\quad \frac{n_1}{F} - 0 = \frac{n_2-n_1}{R_1} + \frac{n_1-n_2}{R_2}$

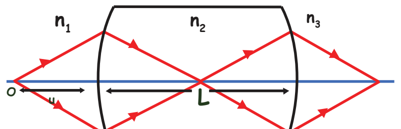
$$\frac{1}{F} = \left[\frac{n_2}{n_1} - 1\right]\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

$$\frac{1}{F} = \left[{}_1n_2 - 1\right]\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

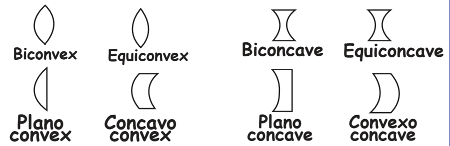
LENSES

Types

Thick lens - Two surfaces are at some distance apart.
Thin lens - Two surfaces are close

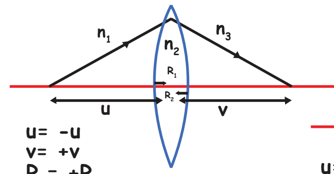


Thin Lens
- Convex
  - Biconvex
  - Equiconvex
  - Plano convex
  - Concavo convex
- Concave
  - Biconcave
  - Equiconcave
  - Plano concave
  - Convexo concave



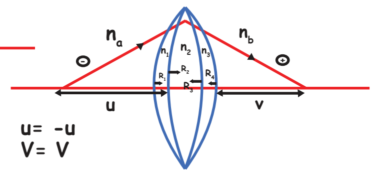
Convex mirror, Convex lens } f=+ve
Concave mirror, Concave lens } f=-ve

IOS METHOD



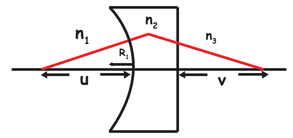
u= -u
v= +v
$R_1 = +R_1$
$R_2 = -R_2$
I-O=S

$$\frac{n_3}{v} - \frac{n_1}{-u} = \frac{n_2-n_1}{R_1} + \frac{n_3-n_2}{-R_2}$$



u= -u
V= V
$R = +R_1, +R_2, -R_3, -R_4$

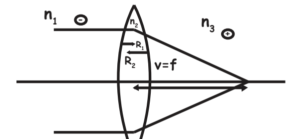
$$\frac{n_b}{v} - \frac{n_a}{-u} = \frac{n_1-n_a}{R_1} + \frac{n_2-n_1}{R_2} + \frac{n_3-n_2}{-R_3} + \frac{n_b-n_3}{-R_4}$$



u= -ve
V= V
R_1= -ve =$-R_1$
$R_2 = \infty$
I-O=S

$$\frac{n_3}{v} - \frac{n_1}{-u} = \frac{n_2-n_1}{-R_1}$$

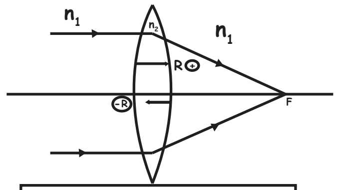
IOS METHOD FOR FOCAL LENGTH



u= ∞
v= f
$R = +R_1, -R_2$

$$\frac{n_3}{f} = \frac{n_2-n_1}{R_1} + \frac{n_3-n_2}{-R_2}$$

EQUICONVEX LENS

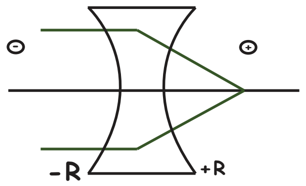


$$F = \frac{R}{2\left[{}_1n_2 - 1\right]}$$

If Prism is placed in air
$$F = \frac{R}{2\left[\frac{n}{1} - 1\right]} = \frac{R}{2\left[{}_1n_2 - 1\right]}$$

In water,
$$F = \frac{R}{2\left[\frac{n_2}{n_w} - 1\right]}$$
⇒ F↑ than air

FOCAL LENGTH OF EQUICONCAVE LENS IN AIR



$$f = \frac{-R}{2\left[{}_1n_2 - 1\right]}$$

PLANO CONVEX LENS

F of planoconvex lens
= 2 x f of equiconvex lens

$$f = \frac{R}{{}_1n_2 - 1}$$

PLANO CONCAVE LENS

Equiconvex = f ⟶ Equi Concave = - f
⇓ ⇓
Plano Convex = 2f Plano Concave = -2 f

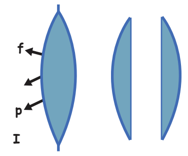
$$f = -\frac{R}{{}_1n_2 - 1}$$

POWER

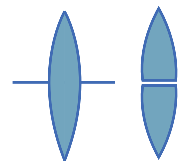
$$\text{Power} = \frac{1}{\text{F in metre}}$$
[Power= 1 dioptre = $1m^{-1}$]

In centimetre, $P = \dfrac{100}{\text{f in cm}}$

CUTTING OF LENS

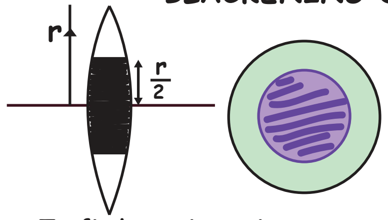


	Before cutting	After cutting
Focal Length	f	2f
Power	P	P/2
Area	A	A
Intensity	I	I



	Before cutting	After cutting
Focal Length	f	f
Power	P	P
Area	A	A/2
Intensity	I	I/2

BLACKENING OF LENS



Intensity ∝ Area of transmission of light
$I \propto A_T$

Blackening of lens ⇒ A_T↓ ⇒ I↓

To find new intensity :

i) Find total A_T (Area of lens before blackening)
ii) Find new $A_T = (A_{total} - A_{opaque})$
iii) $I \propto$ original A_T (Total A_T)

$I' \propto$ new A_T

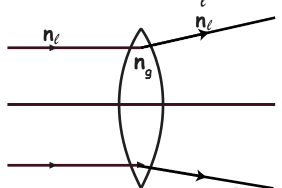
Taking ratio of these two equations we can find I'

Comparison of focal length in air & Liquid

1) If $n_{air} < n_{liquid} < n_{glass}$
- Nature of lens remains same
- Focal length increases

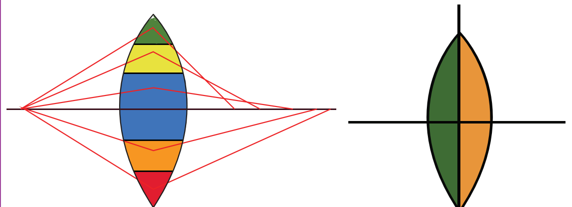
2) Same refractive index $[n_l = n_{glass}]$
- Lens become invisible

3) $n_\ell > n_g$ $\quad f = \dfrac{R}{2\left(\frac{n_g}{n_\ell} - 1\right)}$ =negative



Convex become concave, Concave become convex } Nature of lens changes

NUMBER OF IMAGES FORMED IN MULTIPLE MEDIUM LENS



Number of medium =Number of images

Number of images = 1

LENS FORMULA

To find v when f and u are given

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \qquad \frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{u+f}{uf} \qquad v = \frac{uf}{u+f}$$

MAGNIFICATION

To find size of image

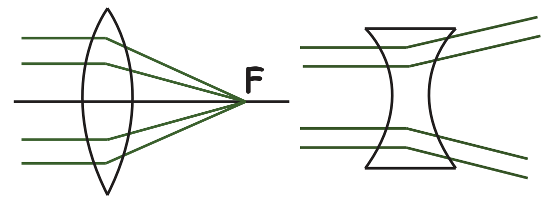
$$m = \frac{[I]}{[O]} = \frac{v}{u}$$

$$m = \frac{f}{f+u}$$
$$m = \frac{f-v}{f}$$

m_T:
- $|m_T| = 1$ ⇒ Same size
- $|m_T| < 1$ ⇒ Small
- $|m_T| > 1$ ⇒ magnified

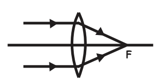
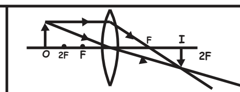
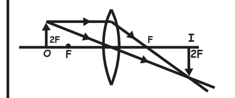
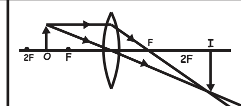
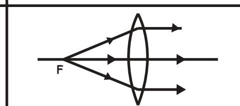
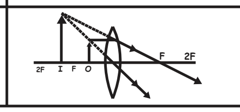
m_T:
- +ve ⇒ Virtual image, Erect
- -ve ⇒ Real image, Inverted

IMAGE FORMATION BY CONVEX AND CONCAVE LENSES

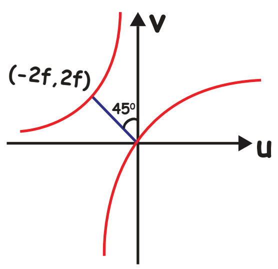


Convex lens
- Converging lens
- Similar to concave mirror

Concave lens
- Diverging lens
- Similar to convex mirror

Sl NO	Position of the object	Ray Diagram	Position of image	Nature of Image
1	At $\lvert u\rvert = \infty$		At F	Real, inverted, diminished
2	Beyond 2F $\lvert u\rvert > 2\lvert f\rvert$		Between F and 2F	Real, inverted, diminished
3	At 2F $\lvert u\rvert = 2\lvert f\rvert$		At 2F	Real, inverted, same size
4	Between F & 2F $f < \lvert u\rvert < 2f$		Beyond 2F	Real, inverted, enlarged
5	At F $\lvert u\rvert = \lvert f\rvert$		At ∞	Cannot be defined
6	within F $\lvert u\rvert < \lvert f\rvert$		On the side of object	Virtual, erect, enlarged

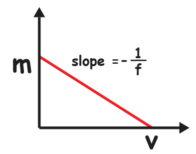
U - V GRAPH



MAGNIFICATION VS V GRAPH

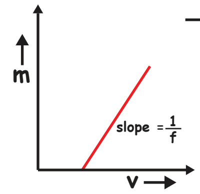
Concave lens

$$m = \frac{f-v}{f}$$
$$m = 1 - \frac{v}{f} = -\frac{v}{f} + 1$$

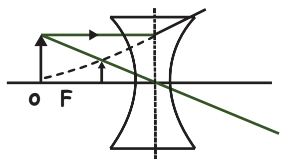


Convex lens

$$m = \frac{v}{f} + 1$$

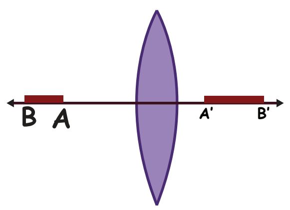


Concave lens



-Real object
-Virtual image
-Formed between F and Pole

AXIAL MAGNIFICATION



$$m_L = \frac{\text{length of image}}{\text{length of object}} = \frac{A'B'}{AB} = \frac{V_A - V_B}{U_A - U_B}$$

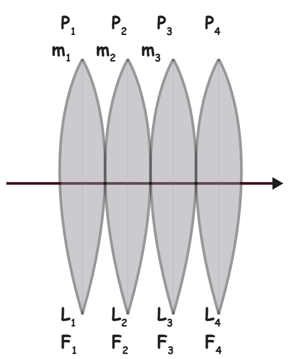
For short object
$$m_L = \frac{dv}{du} = m_T^2$$

PHYSICS WALLAH

RAY OPTICS -2

COMBINATION OF LENSES

Lenses are combined such that there is no gap between them



Power

$P_{comb} = P_1 + P_2 + P_3 + ...$

Focal length

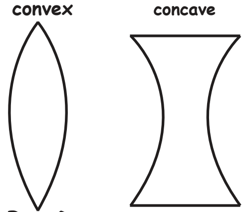
$\frac{1}{f_{comb}} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} + ...$

Magnification ,

$m_T = m_1 \times m_2 \times m_3$ .....

Note :

P_{Comb} determines whether the combination act as converging lens or diverging lens.



P=xD P=-yD

System acts as converging lens if total power > 0

$P_T > 0$

$\Rightarrow P_{convex} > P_{concave}$

$\Rightarrow f_{concave} > f_{convex}$

System acts as diverging lens if total power < 0

$P_T < 0$

$\Rightarrow P_{concave} > P_{convex}$

$\Rightarrow f_{convex} > f_{concave}$

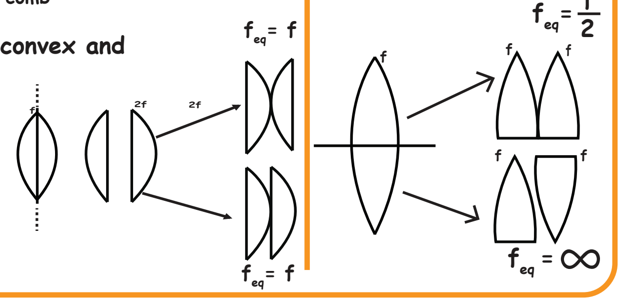
System acts as plane lens /glass if $P_{comb} = 0$

$P_{concave} = P_{convex} = 0$

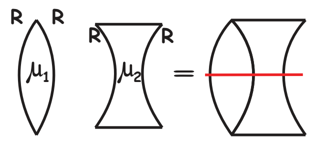
$f_{concave} = f_{convex} = f_{comb} = \infty$

For a combination of convex and concave lenses if $P_{convex} > P_{concave}$

→ combination acts as convex lens



F_{EQ} USING LENS MAKERS FORMULA



$$f_{eq} = \frac{R}{2[\mu_1 - \mu_2]}$$

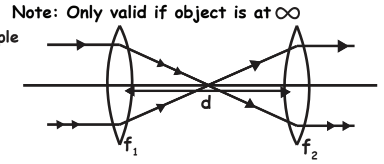
LENSES COMBINED SUCH THAT THERE IS A GAP BETWEEN THEM

Power : $P_{comb} = P_1 + P_2 - dP_1P_2$

$$\frac{1}{f_{comb}} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$$

Note: Only valid if object is at ∞

Example



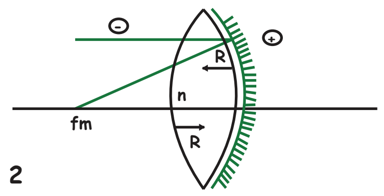
If $P_{comb} = 0$

$$\frac{1}{f_{comb}} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2} = 0$$

$\boxed{d = f_1 + f_2}$

NATURE OF MIRROR IS DETERMINED BY FOCAL LENGTH

To find focal length, split the silvered lens into a lens and a mirror and apply IOS



$u = \infty$

$v = f_m$

$R_1 = +R$

$R_2 = -R$

I - O = S

$$\frac{-1}{f_m} = \frac{n-1}{R} \times 2 + \left[\frac{0+n}{+R}\right] \times 2$$

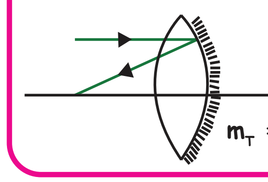
[Ray passes through the surface 2 times thus multiplied by 2]

$$f_m = \frac{-R}{4n-2} = \frac{-R}{2(2n-1)}$$

When equiconvex lens is silvered (mirrored) its focal length become negative (concave mirror) with magnitude

$$f_m = \frac{-R}{2(2n-1)}$$

SILVERING OF LENS



Lens → Mirror

$$m_T = \frac{v}{u} = \frac{[I]}{[o]} = \frac{f}{f-u} = \frac{f-v}{f}$$

SILVERING OF PLANO-CONVEX LENS

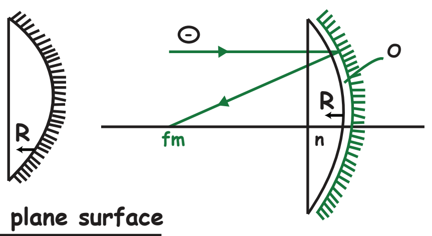
2 possibilities

i) silvering curved surface

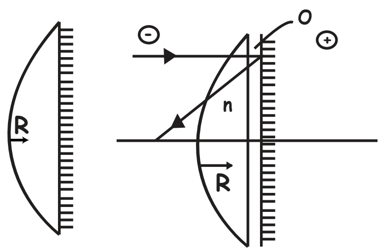
ii) Silvering plane surface

Silvering curved surface

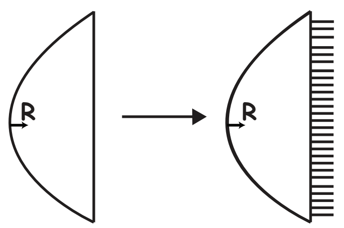
$$f_m = \frac{-R}{2n}$$



Silvering plane surface



$$f_m = \frac{-R}{2(2n-1)}$$



Before silvering

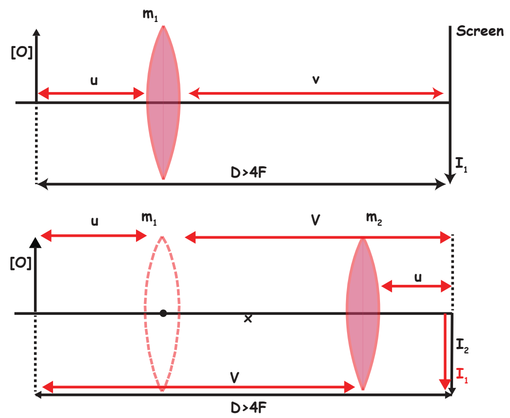
$$f = \frac{R}{(n-1)}$$

After silvering

$$f_m = \frac{-R}{2(n-1)} = \frac{f}{2}$$

LENS DISPLACEMENT METHOD

Distance between object & image > 4F



In lens Displacement method

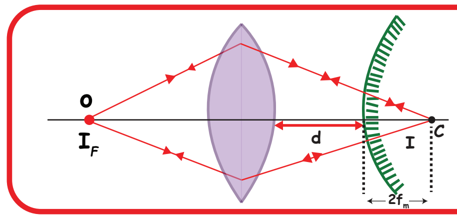
1) $D \geq 4F$

2) $F = \frac{D^2 - x^2}{4D}$ x→distance between 2 position of lens

3) $F = \frac{x}{m_1 - m_2}$ F= Focal length of lens

4) $m_1\ m_2 = 1$

$[O] = \sqrt{I_1 I_2}$



If one is converging & other is diverging $d = v - 2f_m$

LENS MIRROR COMBINATION

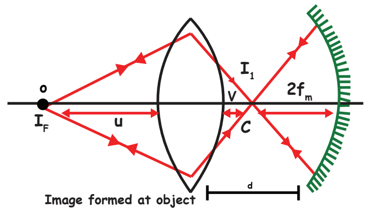


Image formed at object

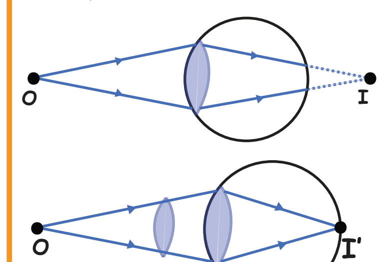
If both lens and mirror are converging. $d = v + 2f_m$

HUMAN EYE

Least distance of distinct vision is 25 cm

Defects of vision and their correction

i) Hypermetropia



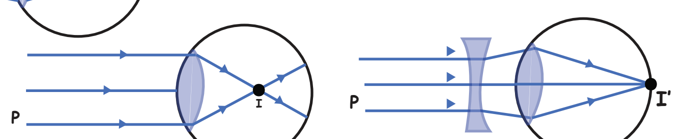
- Long sightedness –Cannot see nearby objects
- Eye focuses incoming light from nearby objects at a point behind retina.
- Correction : Convex lens

F = +ve , Power P= +ve

Focal length = Dd/(D-d)

D→Least distance of distinct vision.

ii) Myopia



- Short sightedness -Cannot see faraway objects
- Light from a distant object arriving at the eye lens may get converged at a point in front of the retina.
- Correction : Concave lens

F = -ve , P = - ve , F = - d , $P = \frac{1}{F}$

iii) Astigmatism

eye cannot focus in horizontal and vertical planes simultaneously

- Correction : Cylindrical lens

iv) Presbyopia

eye suffers both myopia and hypermetropia

- Correction : Bifocal lens

OPTICAL INSTRUMENTS

Simple Microscope

Only one lens [convex lens]

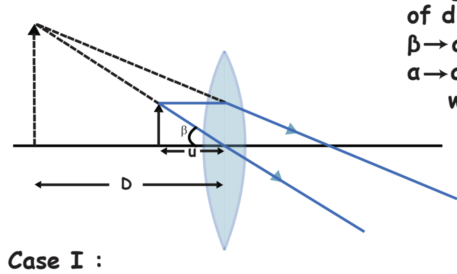


Image is formed at least distance of distinct vision (D)

β→angle subtended by image at eye.

α→angle subtended by object at eye when placed at distance D.

Case I :

Eye under relaxed state or normal vision

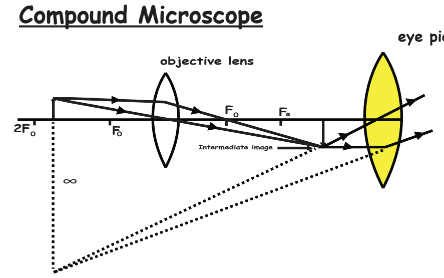
Final image at infinity

Object at f→ $u = u_{max}$ $\boxed{m_{min} = \frac{D}{u_{max}} = \frac{D}{f}}$

Case 2 : Eye under strain

Final image at D $u = u_{min}$ $\boxed{m_{max} = \frac{D}{u_{min}} = \frac{1+D}{f}}$

Compound Microscope



-Has 2 lenses

- Length of microscope
= Distance between lenses.

-Magnification $m = m_o \times m_e$

$[m_e$ – same as simple microscope]

To get magnified image, object placed between F_o & $2F_o$ of objective lens. This image is called intermediate image.

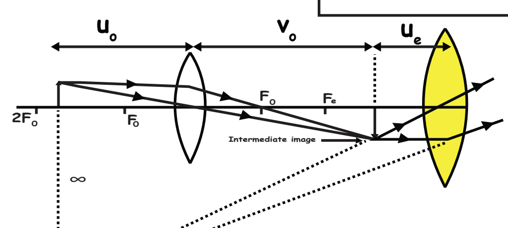
Intermediate image→real, inverted , magnified.

Intermediate image is formed within or at the focus of eyepiece.

Total magnification $m_T = -m_o \times m_e$

m_e same as simple microscope

$$\boxed{m_T = -m_o \times m_e = -\left[\frac{v_o}{u_o} \times \frac{D}{u_e}\right]}$$



Case I :

Eye in relaxed state or final image at ∞, $u_e=f_e$

$m_T=m_{min}=-\left[\frac{V_o}{u_o}\frac{D}{f_e}\right]$ Also $L=L_{max}=V_o+u_{max}=V_o+f_e$

Case 2 : Strained eye

$u=u_{min}$

$L=L_{min}=L_D=V_o+U_e=V_o+u_{min}=V_o+\frac{Df_e}{D+f_e}$

$\boxed{m_D=m_{max}=\frac{-v_o}{u_o}\left[1+\frac{D}{f_e}\right]}$ $\quad \frac{m_{max}}{m_{min}}=\frac{D+f_e}{D}$

$$\boxed{\begin{array}{l} m_D=\left[\frac{L}{f_o}\right]\left[1+\frac{D}{f_e}\right] \\ m_\infty=\left[\frac{L}{f_o}\frac{D}{f_e}\right] \end{array}}$$

Note :

$L_\infty=V_O+U_{max}$ $\quad L_\infty=V_O+f_e$

$L_\infty=\frac{U_of_o}{U_o-f_o}+f_e$ $\quad L_D=V_O+U_{min}$

$L_D=V_O+\frac{Df_e}{D+f_e}$ and $L_D=\frac{U\,f_O}{U_O-f_O}+\frac{Df_e}{D+f_e}$

For microscope, eyepiece larger than objective

$f_e\uparrow\ \Rightarrow m\uparrow$ $\quad f_e\downarrow\ \Rightarrow m\downarrow$

For telescope, eyepiece smaller than objective to increase magnification.

Telescope

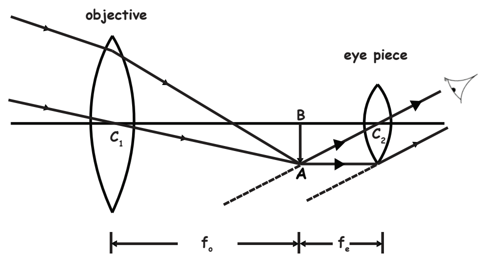
Magnification $m=\frac{f_o}{u_e}$ $\quad u=u_{max}\Rightarrow m=m_{min}$, $\quad u=u_{min}\Rightarrow m=m_{max}$

Length of telescope $L=f_0+u_e$

Normal adjustment /Relaxed eye/final image at ∞

$m_{min}=\frac{f_o}{f_e}$ $\quad$ Length $=L_\infty=f_0+u_e=f_0+f_e$

$\boxed{L_\infty=f_o+f_e}$



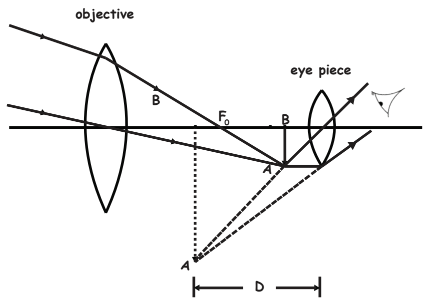
Eye under strain/Final image at least distance of distinct vision.

$u=u_{min}=\frac{Df_e}{D+f_e}$

$m_{max}=\frac{f_o}{f_e}\left[1+\frac{f_e}{D}\right]$

$L_D=f_o+u_e$

$L_D=f_o+\frac{Df_e}{D+f_e}$



Length of Telescope

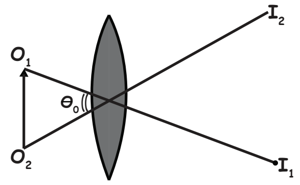
u_e can have values : f_e or $\frac{Df_e}{D+f_e}$

v_o can have values : f_o only

$L_D=v_o+u_e$

RESOLVING POWER

Resolving power$=\frac{1}{\text{Resolving limit}}$



X=Limit of resolution

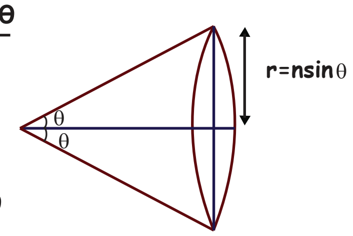
= Resolving limit

MICROSCOPE

Resolving Limit$=\frac{1.22\lambda}{a}=\frac{1.22\lambda}{2n\sin\theta}$, where a = diameter of lens

$R.P=\frac{a}{1.22\lambda}=\frac{2n\sin\theta}{1.22\lambda}$

$R.P\ \alpha\ \frac{1}{\lambda}$



TELESCOPE

$R.P=\frac{a}{1.22\lambda}=\frac{2n\sin\theta}{1.22\lambda}$

RAY OPTICS -2